

Large Eddy Simulation of Active Flow Control on a Circular Cylinder

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INTRODUCTION

Bluff-body flows have diverse engineering applications and have been extensively investigated to reduce aerodynamic drag. The circular cylinder is one such geometry, with flow characterized by boundary-layer separation, shear-layer transition, and periodic vortex shedding. Flow complexity depends on cylinder diameter and the Reynolds number Re_D ; in the subcritical region ($300 < Re_D < 100,000$), flow separation remains laminar before transitioning to turbulent wake flow. The circular cylinders have often been investigated with active flow control (AFC) devices. One of the AFC devices that has displayed significant potential in recent years is the synthetic jet actuator (SJA). An SJA is a zero-net-mass-flux device that adds momentum to the flow through periodic ingestion and expulsion cycles. SJAs function without a fluid reservoir or plumbing, making them ideal for AFC.

OBJECTIVE

- Investigate the effectiveness of a five SJA array in flow control over a circular cylinder at $Re_D = 3900$.
- Characterize and examine the effects of the three-dimensional flow structures generated by the SJA array
- Identify ideal control parameters

METHODS

- Wall-Resolved Large Eddy Simulations (WR-LES)
- Wall-Adapting Local Eddy-Viscosity (WALE) Model
- OpenFOAM V2412
- Analytical pulsing jet Boundary Condition for SJA velocity inlet
- O-Grid for baseline circular cylinder flow

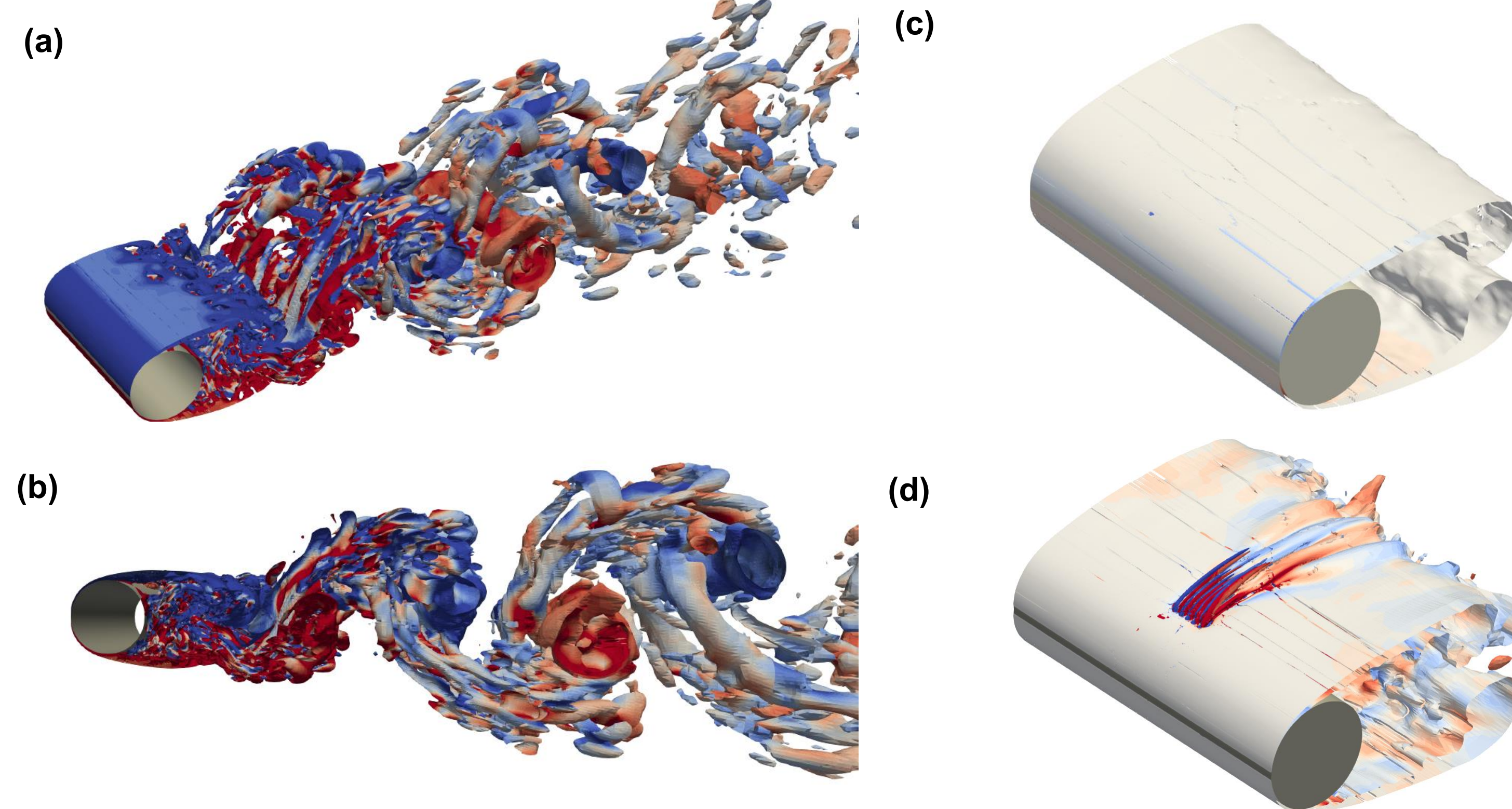
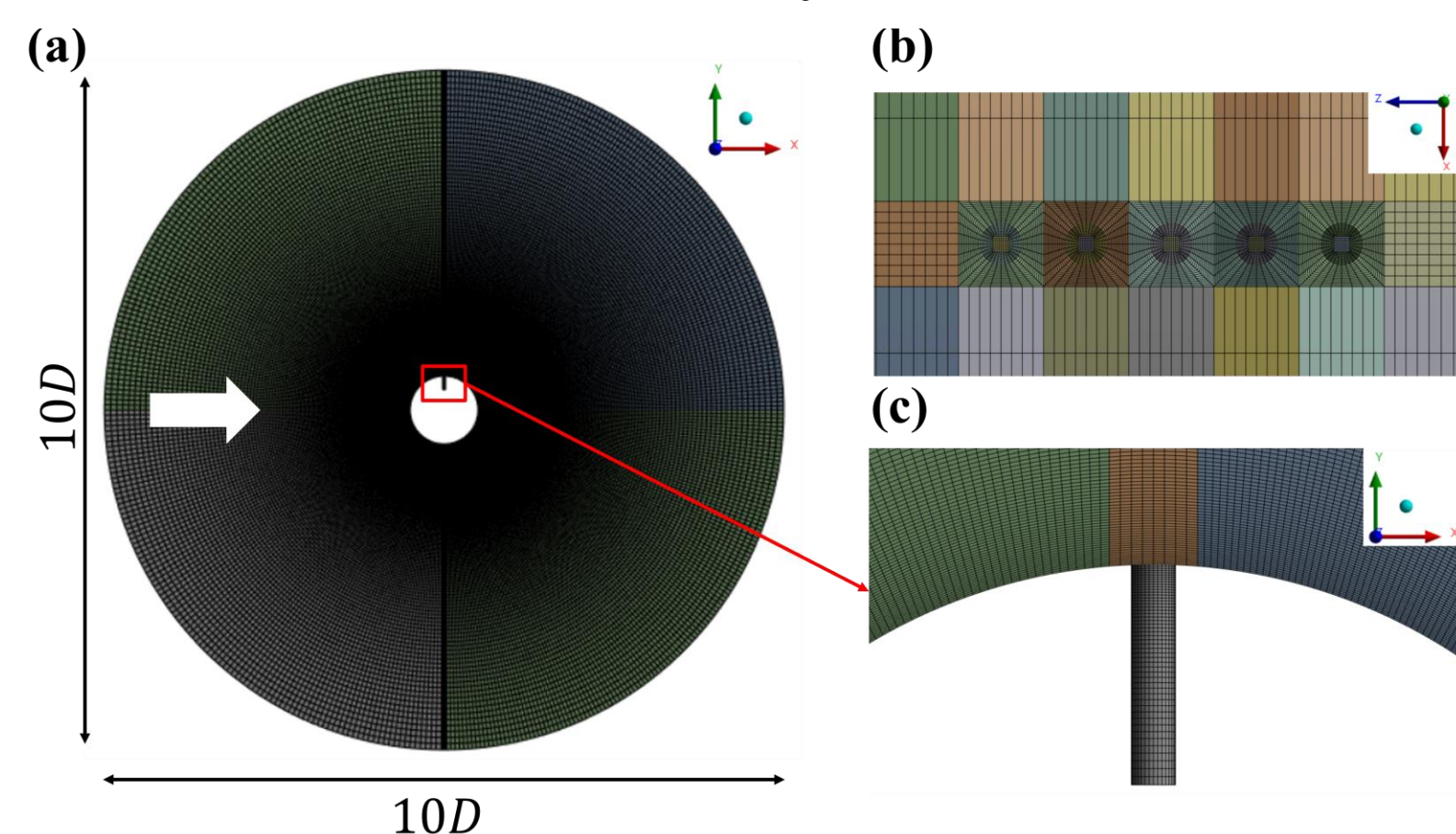


Figure 1. Q-criterion contours of instantaneous (a) baseline isometric view, (b) baseline side view; time-averaged (c) baseline isometric view, (d) actuated isometric view.

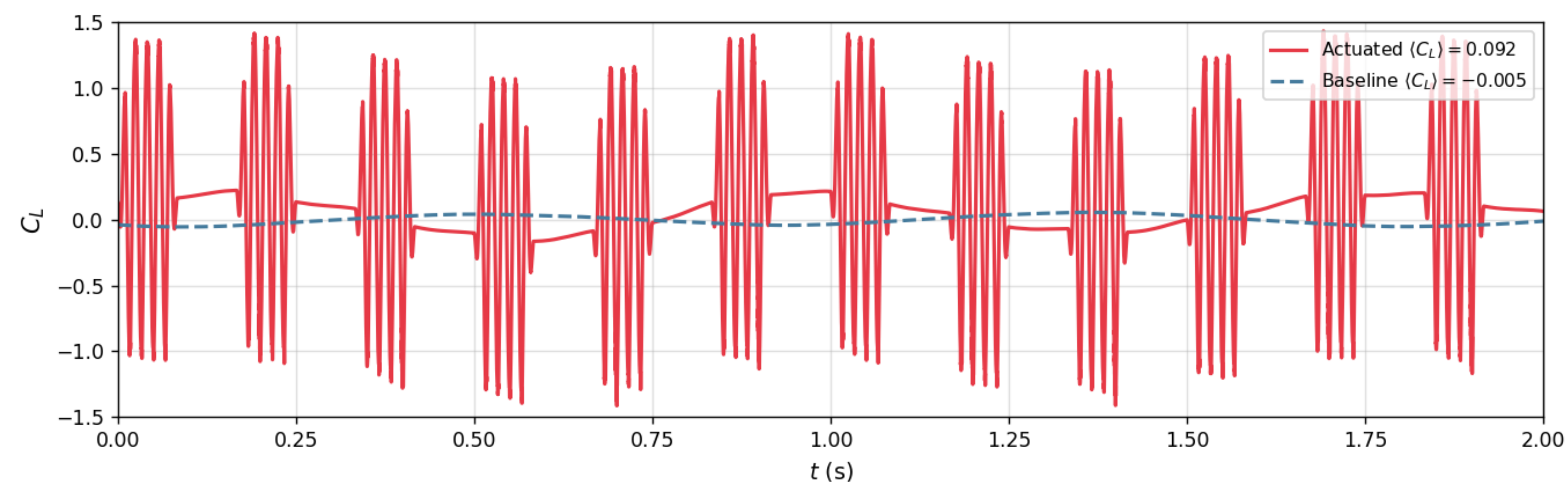


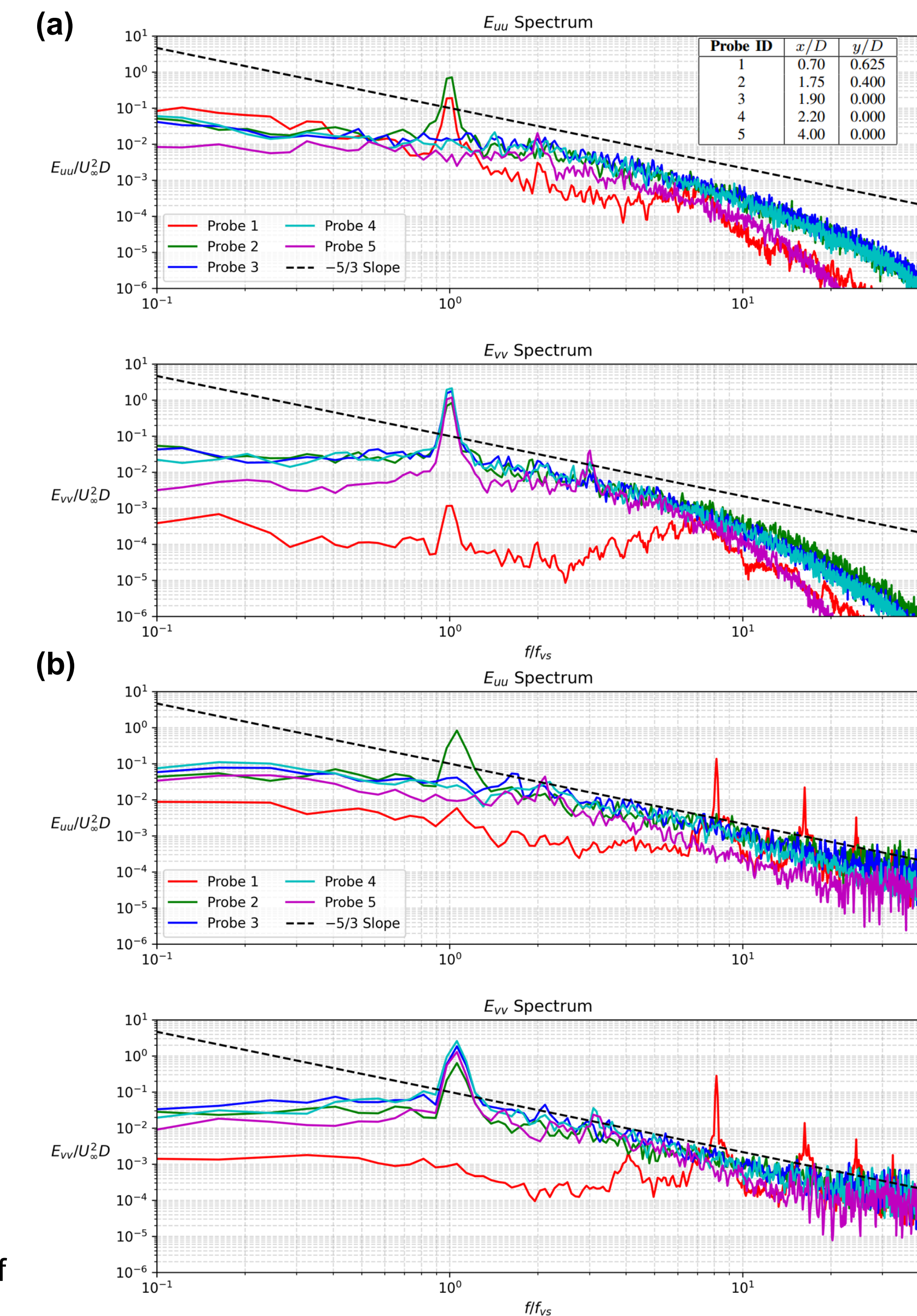
Figure 2. Lift coefficient C_L comparison of baseline (blue) and actuated (red) case, displaying lift generation of the SJA array.

CONCLUSION

- Actuation significantly increases lift coefficient C_L of the cylinder
- The SJA array demonstrated significant control authority despite its limited spanwise coverage
- Spectral analysis reveals that actuation eliminates the baseline shedding frequency peak in most wake regions on the actuated side

RESULTS

Figure 3. Power spectral density (PSD) of streamwise (E_{uu}) and transverse (E_{vv}) velocity fluctuations at the five probe locations. Plots show (a) the baseline unactuated flow and (b) the flow with SJA control.



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